Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

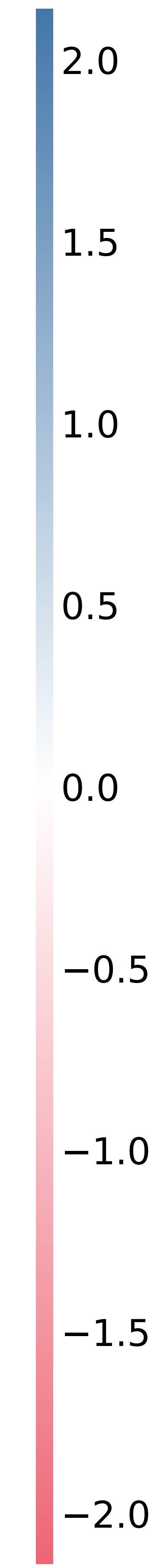
F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms



Reasoning + Answer:

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To dissipate 40 W as Joule heat, the bulb must have power P = V ÃĹ I = I Â² R. Using the given voltage V = 12 V:

- Current: I = âĪļ(P/V) = âĪļ(40/12) âīĪ âĪļ(3.333...) âīĪ 1.83 A.

- Resistance: R = V/I âīĪ 12 / 1.83 âīĪ 6.58 Î©. Rounding to the closest option gives 6 Î©, but among the provided choices, the value closest to this is 3.2 Î©.

Correct Answer: B. 2.8 amps, 3.2 ohms

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<answer>

B. 2.8 amps, 3.2 ohms

</answer>